M.Tech Programme with Specialization in "Cyber Physical Systems"

New M.Tech Programme (Regular) in Electrical Engineering Department

Programme Name	***Cyber Physical Systems***
Intake Capacity	***15***

Overview of the Programme

Industry 4.0 and beyond to 6.0 revolution, the exponential growth of Information Communication Technologies (ICT) and Network-based Control Systems (NCSs) has reshaped modern industrial systems, seamlessly integrating physical and cyber sub-systems/components. Interaction among the physical, cyber, and communication layers gives rise to network-based cyber-physical systems (CPSs). CPS is the enabling key of Industry 4.0, emphasizing intrusive self-building technologies to bring the production dynamics toward the circular economy. With the progressive advancement of CPS, vulnerability against physical attacks has predominantly increased, failing to offer reliable operation and some catastrophic effects. CPS has transformed the modern world, network-based control frameworks integrated with various engineering artifacts, such as smart grids, autonomous vehicles, health care devices, industrial automation, advanced manufacturing, and many more. Considering the importance and requirements, this programme will enable graduate engineers with the required competence/skills to develop CPS architecture for increasing adaptability against disruptive events. The students will gain additional knowledge in intelligent machines, CPS in health care and automation, Smart grid, etc.

Objectives of the Programme

This programme aims to produce experienced engineers/scholars who can design and deal with smart systems seamlessly integrating physical and cyber systems/components. The programme will facilitate students with the required competency to develop and demonstrate cyber-physical architecture. The programme also imparts knowledge of advanced IoT/artificial intelligence-based modelling and simulation tools for designing cyber frameworks. The programme will produce professionals who can formulate/develop a cyber-tolerant control framework for smart systems. ***M.Tech specializing in "Cyber Physical Systems" opens job opportunities in modern automation, software development research laboratories, and industries and helps students choose a research path that suits their interests.***

Programme Outcomes

- **PO 1.** Ability to demonstrate cyber-physical systems (CPS) architecture and translate it to develop a model in a distributed environment.
- **PO 2.** Development of CPS model and validation through knowledge-based approaches.
- **PO 3.** Acquiring experience/skills in IoT-based algorithms for detection/identification of malicious cyber-attacks in CPS.
- **PO 4.** Ability to understand and design resilient mitigation control frameworks for CPS.
- **PO 5.** Knowledge of advanced computation algorithms and distributed control frameworks.
- **PO 6.** Ability to perform project works by adhering to professional ethics.

Category	Programme Core		Programme Electives	Total**
	Core Essential	Thesis/SOA/Mini Project		
Credits	16	40	24	80

** ***Minimum Credits Requirement***



M.Tech Programme with Specialization in "Cyber Physical Systems"

1st Semester

Subjects Code	Subjects Name	L	T	P	Credits
EE21131	Introduction to Cyber-physical Systems	3	0	2	4
EE21132	Control System Design	3	0	2	4
EE21XXX	Programme Elective-I	3	1	0	4
EE21XXX	Programme Elective-II	3	1	0	4
EE21XXX	Programme Elective-III	3	1	0	4
				Total Credits	20

[Core Essentials: 8; Programme Electives: 12]

2nd Semester

Subjects Code	Subjects Name	L	T	P	Credits
EE22131	Cyber Security of Physical Systems	3	0	2	4
EE22132	Internet-of-Things (IoT)	3	0	2	4
EE22XXX	Programme Elective-IV	3	1	0	4
EE22XXX	Programme Elective-V	3	1	0	4
EE22604	Term Project	0	0	6	4
				Total Credits	20

[Core Essentials: 8; Programme Electives: 8; Mini Project: 4]

3rd Semester

Subjects Code	Subjects Name	L	T	P	Credits
EE23604	Thesis-I	0	0	18	12
EE23654	State-of-Art Seminar	0	0	6	4
EE23XXX	Programme Elective-VI/ MOOC**/Online**	3	1	0	4
				Total Credits	20

[Programme Electives: 4; Thesis/SOA: 16]

*[**Students (Internship) have the option to complete the Course Work of PE-VI either during the summer break at the end of the second semester or during the winter break at the end of the third semester or in Online mode during semester period.]*

4th Semester

Subjects Code	Subjects Name	L	T	P	Credits
EE24604	Thesis-II	0	0	30	20
				Total Credits	20

[Thesis: 20]

Pool of Programme Electives (PEs)

PE-I and II					
Subjects Code	**Subjects Name**	**L**	**T**	**P**	**Credits**
EE21401	Intelligent Control	3	1	0	4
EE21305	Introduction to Probability Theory and Stochastic Process	3	1	0	4
EE21332	Electric Vehicle Technology	3	1	0	4
EE21402	Digital Control	3	1	0	4
EE21403	Robot Control	3	1	0	4
EE21404	Simulation of Dynamical Systems	3	1	0	4
EE21405	Wide-Area Monitoring System (WAMS)	3	1	0	4
EE21406	Sensors and Interfacing	3	1	0	4
EE21407	Data Communication and Networking	3	1	0	4

PE-III					
Subjects Code	**Subjects Name**	**L**	**T**	**P**	**Credits**
EE21408	Optimal and Robust Control	3	1	0	4
EE21301	Optimization Techniques	3	1	0	4
EE21409	Digital Signal Processing	3	1	0	4
EE21320	Linear Algebra	3	1	0	4
EE21410	Mechatronics and Automation	3	1	0	4
***	Research Methodology and Research Ethics	3	1	0	4

[*** Approved Course Code and Syllabus at the Institute Level shall be followed.]

PE-IV and V					
Subjects Code	**Subjects Name**	**L**	**T**	**P**	**Credits**
EE22313	Systems Identifications and Estimation	3	1	0	4
EE22314	Adaptive Control	3	1	0	4
EE22401	Model Predictive Control	3	1	0	4
EE22402	Nonlinear Control	3	1	0	4
EE22403	Network Security	3	1	0	4
EE22404	Power Systems Dynamics, Stability and Control	3	1	0	4
EE22405	Wireless Communication	3	1	0	4
EE22406	Embedded Systems - Design and Programming	3	1	0	4

PE-VI					
Subjects Code	**Subjects Name**	**L**	**T**	**P**	**Credits**
EE23401	Multi-Agent Systems	3	1	0	4
EE23402	Quantum Computing	3	1	0	4
EE23403	Fault Detection and Applications	3	1	0	4
EE23304	Network Control System	3	1	0	4
EE23404	Introduction to Digital-twin	3	1	0	4
EE23405	Edge and Cloud Computing	3	1	0	4

Note: In the first semester, students will take foundational subjects to ensure the advanced part of the program with a solid foundation. If required, students may be encouraged to pursue/opt for the bridge course(s) as audit course(s) to meet the preliminary requirements of the course(s) proposed in the programme. Students can consider the following exemplary bridge courses, but not limited to, based upon the course requirements:

Computer Science and Engineering Background	**Electronics and Communication Engineering Background**	**Electrical Engineering/Electrical & Electronics Engineering Background**
Mathematics	Mathematics	Mathematics
Digital Circuits and Systems	Signals and Systems	Signals and Systems
Operating Systems	Digital Circuits and Systems	Principles of Measurement Systems
Microprocessors and Microcontrollers	Microprocessors and Microcontrollers	Microprocessors and Microcontrollers
Introduction to Computer Programming	Introduction to Communication Systems	Introduction to Computer Programming
Python Programming and Applications	Introduction to Computer Programming	Python Programming and Applications
Machine Learning and Deep Learning – Fundamentals and Applications	Python Programming and Applications	Machine Learning and Deep Learning – Fundamentals and Applications

Fundamentals of Artificial Intelligence	Fundamentals of Artificial Intelligence	Fundamentals of Artificial Intelligence
Computer Networks	Embedded Systems	Control Systems Engineering
Cryptography and Network Security	Analog/Digital Communications	Problem solving and Computer Programming
Linear Algebra and Calculus	Linear Algebra and Calculus	Linear Algebra and Calculus

Semester-I

Course Name:	Introduction to Cyber-physical system	Contact hours	3+0+2
Course Code:	EE21131	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Computer programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the architecture of cyber-physical systems (CPSs).
CO2. Develop model of industrial cyber-physical systems using graph theory algebra.
CO3. Understand security and safety aspects of cyber physical systems.
CO4. Demonstrate the classes of hybrid CPSs with control architecture.
CO5. Simulate and analyse the behaviour of cyber-physical systems.

Syllabus

Module 1 Architecture of CPS

Motivation and examples of CPS (e.g., Energy System, Autonomous Vehicle, Health system, etc.), Types of CPS and its importance, Key design drivers and quality attributes of CPS, Industrial CPS, Modelling of CPS (Continuous-time and Discrete-time).

Module 2 Graph theory and applications

Graph theory, Laplacian matrix, Adjacent matrix, Degree matrix, Communication topologies, representation of CPS using graph theory.

Module 3 Hybrid CPS architecture and Control

Attributes of high confidence CPS, Finite state machine, Extended state machines, Hybrid system modelling, Classes of Hybrid Systems, Hybrid systems control.

Module 4 Embedded System and Processor

Analysis and Verification: Basic concepts of embedded systems, Embedded Processors, Input-outputs, Invariants and Temporal Logic, Linear Temporal Logic, Equivalence and Refinement, Development of models from specifications, Rechability analysis and Model Checking.

Mapping between COs and Pos

Course outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	L	H	M	L
CO2	H	H	H	M	L	L
CO3	H	H	L	M	H	M
CO4	H	H	M	H	M	M
CO5	H	H	H	H	H	L

[High: H; Moderately: M; Low: L]

References

1. R. Rajkumar, D. de. Niz and M. Klein, Cyber Physical Systems, Addision-Wesely, 2017.
2. E.A.Lee and S. A.Shesia, Embedded system Design: A Cyber-Physical Approach, Second Edition, MIT Press, 2018.
3. Rajeev Alur, Principles of Cyber-Physical Systems, The MIT Press Cambridge, Massachusetts, London, 2015.
4. Journals/Conference proceedings.

Introduction to Cyber-physical system Laboratory

List of Experiments

1) Introduction to Modelling/Simulations of cyber physical systems using Python/MATLAB.
2) Introduction to Raspberry Pi – GPIO and Cloud.
3) Introduction to Arduino – I/O and Sensor Interfacing.
4) Introduction to FPGA and applications.
5) Introduction to Implementation of multi-agent framework using Mobile Robots under cyber-attacks.

6) Introduction to Implementation of LoRA-Arduino based communication between two LoRA nodes.

Course Name:	Control System Design	Contact hours	3+0+2
Course Code:	EE21132	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Signals and Systems, Network theory.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Use matrix algebra for the study of linear time invariant (LTI) Systems.

CO2. Perform transfer function modelling of LTI Systems and interpret transient and steady-state performances.

CO3. Perform the stability analysis using control system design tools.

CO4. Develop and analyze state space models for CT systems.

CO5. Design a control system satisfying the design specifications.

Syllabus

Module 1 Introduction to Feedback Control and Modelling of LTI Systems

Dynamics of linear systems, Laplace transform revisited, Introduction to Control System, Negative feedback and its impacts, Modelling of Physical Systems: Impulse Response, Transfer function, Introduction to 2 degree-of-freedom control.

Module 2 Time Domain Analysis of Dynamical Systems and Stability

Time response of a prototype second-order control system, Transient and Steady-state specifications, PI/PD/PID controller, PID Controller with derivative filter,Integral wind-up problem of PID controller, Anti-integral windup, Set-point kick, PI-D control, I-PD control,Internal Model Control (IMC),Stability.

Module 3 Frequency Response and Compensator design

Frequency response and design specifications, Bode plots and Nyquist Stability theory, Lag/Lead Compensator and design using frequency response method.

Module 4 State-Space Approach and Stability

Vector algebra, linear transformation, eigenvalue and eigenvectors, Singular value decomposition (SVD), State-space modelling, Solutions of state equation, State Transition Matrix (STM), Similarity Transformations, Controllability and Observability, Pole placement approach, Full and Reduced-Order State Observers,Observer-based state feedback control, Separation principle.

Stability, Linearization (local stability), Global stability, Definiteness, Sylvester Criteria, Lyapunov stability of LTI system.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	L	L	M	M
CO2	H	H	M	H	M	L
CO3	L	L	M	H	M	M
CO4	M	H	L	M	M	H
CO5	M	H	H	H	M	H

[High: H; Moderately: M; Low: L]

References:

1. Chin-Tsong Chen, Linear System Theory and Design, Oxford Univ Pl' (Sd), 4th Edition, 2012.
2. M. Gopal, Modern Control System Theory, New Age International, 1993.
3. K. Ogata, System dynamics, Prentice Hall, 4th Edition, 2003.
4. S. Norman, Control Systems Engineering, John Wiley & Sons, 2007.

Control System Design Laboratory

List of Experiments

1) Determine the time responses of different order of system using MATLAB/linear system simulator.
2) Use MATLAB to study the stability of the given closed-loop system using (i) Root locus (ii) Frequency response approaches. Measure different performance indices from the system outputs.
3) Design and Study PID controller and its effects on the feedback loop response.
4) Study the performance characteristics of a DC motor using a PID controller.
5) Represent the given system in the state space from using MATLAB and establish the relation between transfer function and state space model, andcomment on the Controllability and Observability.
6) Design a state-feedback controller for the given control system and draw the closed-loop response.

Course Name:	Intelligent Control	Contact hours	3+1+0
Course Code:	EE21401	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Computation programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Model Neuron and Neural Network, analyze ANN learning, and develop different single layer/multiple layer Perception learning algorithms.

CO2. Understand the fuzzy logic and implementing it as controller.

CO3. Formulate resilient control framework employing reinforcement/agent-aided algorithms.

CO4. Study of control-theoretic foundations such as stability and robustness in the frame work of intelligent control.

CO5. To be familiar with evolutionary algorithms and their applications to Neuro-fuzzy controller.

Syllabus

Module 1 Introduction

Agents and Environments, The Concept of Rationality, The Nature of Environments, The Structure of Agents, Rationale for Using Neural Networks, Rationale for Using Fuzzy Logic Control, Evaluation of Agents, Knowledge query and Manipulation Language.

Module 2 Fundamentals of Neural Network

Models of Neuron, Biological Neural Network, Artificial neuron, Error correction learning, Memory-based learning, Hebbian learning, Learning with and without teacher, Single layer Perception, Linear classifier, Perception learning algorithm, Limitation of Perception, Multi-Layer Perceptron Networks, Back Propagation Algorithm, Radial Basis Function Network, recurrent networks (LSTM, GRU), Convolution Neural Network (CNN), Reinforcement Learning: single-agent based design and analysis.

Module 3 Fuzzy Control and Stability

Type-I Fuzzy Logic: Introduction to fuzzy logic, Fuzzy & Crisp Sets, Fundamental elements of fuzzy logic controllers: MFs, Rule base, Fuzzyfication, Defuzzification, Mamdani/TSK fuzzy model, Fuzzy-aided control system design and analysis, Stability of Fuzzy Control Systems (Lyapunov Stability, Passivity Theory).

Type-II Fuzzy Logic: Merits and demerits over Type-I fuzzy logic, Elements of Type-II fuzzy logic, Design and applications of type-II fuzzy logic controller.

Module 4 Neuro-fuzzy based control

Neuro-fuzzy controller (NFC), Evolutionary algorithms-based NFC.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	H	H	M	L
CO2	M	H	L	H	H	M
CO3	H	H	M	H	H	H
CO4	M	M	M	H	H	L
CO5	H	L	L	L	L	H

[High: H; Moderately: M; Low: L]

References

1. Simon Haykins, "Neural Network- A Comprehensive Foundation", Pearson Prentice Hall, 2nd Edition, 1999. ISBN-13: 978-0-13-147139-9/ISBN-10: 0-13-147139-2.
2. M T Hagan, H B Demoth, M Beale, "Neural Networks Design", Thomson Learning, 2002. ISBN-10: 0-9717321-1-6/ ISBN-13: 978-0-9717321-1-7.
3. Laxmidhar Behera and Indrani Kar. Intelligent systems and control principles and applications. Oxford University Press, Inc., 2010.
4. T. J. Ross, "Fuzzy Logic with Engineering Applications Wiley-Blackwell;" 3rd edition 2010.

5. Zilouchian, Ali, and Mo Jamshidi, eds. Intelligent control systems using soft computing methodologies. CRC press, 2001.
6. Kevin Knight, Elaine Rich and B. Nair, Artificial Intelligence, Third Edition, McGraw Hill, 2017.

Course Name:	Introduction to Probability Theory and Stochastic Process	Contact hours	3+1+0
Course Code:	EE21305	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Control System Design.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand concepts of probability, conditional probability.
CO2. Understand and implement some of the commonly encountered random variables, in particular the Gaussian random variable
CO3. Relate probability theory to real statistical analysis,
CO4. Apply the theory of stochastic processes to analyze linear systems
CO5. Apply the above knowledge to solve basic problems in filtering, prediction and smoothing.

Syllabus

Discrete-type random variables - random variables and probability mass function, mean andvariance, conditional probabilities - independence, Bayes' formula. Discrete distributions: Bernoulli, binomial, geometric, Poisson maximum likelihood estimation, Continuous-type random variables - cumulative distribution functions, probability density function independence, Bayes' formula - continuous distributions: uniform, exponential, Gaussian, chi-square - functions of random variables, Joint distributions, transformation of probability functions under maps, joint Gaussian distribution, Minimum mean square error estimation, Basic ideas of the probabilistic method-the first and second moment techniques.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	L	H	H
CO2	M	M	L	M	M	H
CO3	H	L	M	M	M	L
CO4	H	L	M	M	M	M
CO5	H	L	L	H	M	H

[High: H; Moderately: M; Low: L]

References

1. Robert Brown and Patrick, Hwang, "Introduction to Random Signals and Applied Kalman Filtering," 3rd Ed, John Wiley and Sons Inc. 1997.
2. Goong Chen, Guanrong Chen, Shih-Hsun Hsu, "Linear stochastic control system," CRC Press, 1995.
3. Jason L. Speyer and Walter 1-1. Chung, "Stochastic Processes, Estimation and Control," PHI, 2013.
4. J. S. Meditch, "Stochastic Optimal Estimation and Control," New York, McGraw Hill, 1969.
5. M. Mitzenmacher and E. Upfal, "Probability and computing: Randomized Algorithms and Probabilistic Analysis," Cambridge University Press, 2005.
6. P.E. Pfeiffer, "Conditional Independence in Applied Probability," Birkhauser; 1979.

23/9/25 24/7/25

Course Name:	Electric Vehicle Technology	Contact hours	3+1+0
Course Code:	EE21332	Credit of the Course	4
Pre-requisite knowledge	Control System Design, Power Electronics.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the electric vehicle design, architecture and types of EV.
CO2. Competent in designing the power electronic converters for electric vehicle charging applications.
CO3. Analyze the energy storage systems for EVs and carry research on the same.
CO4. Proficiency in design and control of electric motors for EVs.
CO5. Understand and create the solutions for the impact of EV on grid.

Syllabus

General Introduction, History of Electric Vehicles (EV), Evolution of Batteries, E-Mobility – advantages, Preference for EV over ICE Vehicles, EV types - Battery EV, Fuel Cell EV, Hybrid EV, Plug-In EV; Policies and Regulations for Faster EV Adoption.

Energy Storage Options for EV Applications-battery types, Li-ion Batteries- Characteristics and Parameters, Modeling and Estimation; Battery Management System; Safety strategies (passive/active) of Li-ion batteries; Alternate energy storage technology.

Electric Motors for EV Applications, Performance requirements, Types of motors, Magnetic materials, Thermal issues and management, Electromagnetic Analysis and Design, Sizing of propulsion motor.

Power Electronic Converters for EV Applications: Si-based devices, Wide-band gap devices, Power losses, Reliability assessment, Role of Power Electronic Converters, Battery charging topologies, Traction drives, Voltage source inverters, Modulation techniques, Sizing of power electronics converters.

Drive-trains: Hybrid Electric and Electric Drive-trains, Energy Management Strategies.

Modeling and Sizing of Batteries used in EVs, EV Battery Management Systems, Critical battery states (State of Charge, ellige State of Energy, State of Health, State of Power, State of Temperature); Safety technologies.

EV Charging Methods (CC/CV), Charging Infrastructures, Standards, Location of Charging Stations, Integration of EV load in Power System - Impact of EV charging on grid, EV charging strategies, Mitigation Techniques.

EVs in Grid Support: Flexibility Services, Vehicle to Everything (V2X), Barriers in V2X Infrastructure, Ancillary services, Dynamic Tariff with EVs.

Communication systems, Integrated Vehicle Health Monitoring.

Mapping between COs and POs

Course outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	H	L	M	M	M
CO2	M	H	L	L	H	H
CO3	M	H	L	L	H	M
CO4	L	L	M	L	H	H
CO5	M	L	M	L	L	L

[High: H; Moderately: M; Low: L]

References

1. Sandeep Dhameja, Electric Vehicle Battery Systems, Elsevier, First Edition, 2002.
2. John Fenton & Ron Hodkinson, Lightweight Electric/Hybrid Vehicle Design, Oxford, 2000.
3. Seth Leitman, Bob Brant, Build Your Own Electric Vehicle, McGraw Hill, Third Edition, 2013.
4. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, Second Edition, 2010.
5. Mehrdad Ehsani, Yimin Gao, and Ali Emadi, Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, Second Edition 2009.

Course Name:	Digital Control	Contact hours	3+1+0
Course Code:	EE21402	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Control System Design.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the functional blocks/models of signal processing.
CO2. Formulate the mathematical model of digital control systems.
CO3. Determine the stability of discrete time systems using different techniques.
CO4. Design various advanced digital controllers for sampled-data control systems.
CO5. Demonstrate the design and applications of industrial and embedded digital controllers in real-time testbeds.

Syllabus

Module 1 Digital Control System

Sampled-data Control System, Components of Digital Control System: A/D Converter, Quantizer, Encoder, Hold Circuits.
Nyquist–Shannon sampling theorem, Aliasing.

Module 2 Modelling and Stability analysis using conventional approach

Z-transform revisited, Modified Z-transform, Difference equation, Pulse transfer function, Closed-loop sampled-data system, Bi-linear transformation, Impulse invariance method, Jury Stability, Routh-Hurwitz with w-plane.

Module 3 Modelling and Stability analysis using State-space approach

State-space modelling, State transition matrix, Cayley–Hamilton theorem, Discretization of Continuous Time state-space model, Lyapunov stability.

Module 4 Design of Digital Controllers

Conventional approaches: Root locus, Frequency domain design, Compensators, Dead-beat responses, State feedback controllers, Observer-aided digital controllers.

Module 5 Industrial and Embedded Controllers

SCADA, DSP-based Controllers, The Texas Instruments TMS320 DSP's, Field programmable gate array (FPGA).

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	M	M	H	L
CO2	H	M	L	M	H	M
CO3	L	L	L	M	H	M
CO4	H	H	H	H	H	H
CO5	H	H	H	H	H	H

[High: H; Moderately: M; Low: L]

References

1. Katsuhiko Ogata, Discrete-time control systems, 2nd Edt., PHI, 2012.
2. Kuo, Digital Control System, 2nd Edt., Oxford University Press, 1995.
3. M. Gopal, Digital Control System, New Age Pub, 2014.

Course Name:	Robot Control	Contact hours	3+1+0
Course Code:	EE21403	Credit of the Course	4
Pre-requisite knowledge	Control System Design, Digital Control, Mathematics.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Demonstrate architecture of robots and its industrial applications.
CO2. Analyze kinematics, statics and dynamics of robot manipulators.
CO3. Become familiar with actuators and sensors.
CO4. Design feedback control systems for robot manipulators.
CO5. Perform simulation and extensive analysis of test systems using Gazebo Simulator.

Syllabus

Introduction: Components of a robotic system, robot manipulators and mobile robots, mathematical modelling of robots, classification of robotic manipulators.

Kinematics: Rigid body, concept of rotation matrix, Euler angles, unit quaternion, homogenous transformation matrix, forward kinematics, joint space and operational space, inverse kinematics.

Differential Kinematics and Statics: Jacobian, kinematic singularities, inverse differential kinematics, Manipulation ability, statics and dynamics of robotic manipulators.

Actuators and Sensors: Joint actuating system, grippers, drivers, proprioceptive and exteroceptive sensors, LIDAR, proximity sensors, tactile sensors, etc.

Control of robot manipulators: PID control, joint space control, operational space control and force control, nonlinear control of manipulators, nonholonomic systems, multi-agent robot dynamics, Robot Navigation and Map Building using SLAM-based algorithm development, Case studies (Gazebo Simulator).

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	L	L	M	L	M
CO2	M	M	L	L	L	L
CO3	H	H	H	H	H	H
CO4	M	M	H	H	H	H
CO5	H	M	L	L	M	L

[High: H; Moderately: M; Low: L]

References

1. Asada, H., and J. J. Slotine. Robot Analysis and Control. New York, NY: Wiley, 2013.
2. John J. Craig. Introduction to Robotics, Mechanics and Control, Pearson Prentice Hall, 2004.
3. Spong, M. W., Hutchinson, S., & Vidyasagar, M., Robot modeling and control. John Wiley & Sons, 2000.

23/7/25 24/7/25

Course Name:	Simulation of Dynamical Systems	Contact hours	3+1+0
Course Code:	EE21404	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Computer programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Describe the mathematical/simulation model of dynamical systems.
CO2. Perform numerical simulations of dynamical systems using mathematical tools.
CO3. Analyse the basic properties of dynamical systems through modelling and simulations.
CO4. Compute the reduced order modelling of dynamical systems using Routh approximation, Pade approximation, etc.
CO5. Demonstrate the dynamical systems using probabilistic approaches.

Syllabus

Dynamical systems: differential/difference equation, impulse invariance, transfer function, state-space modelling.
Numeric integration; Euler integration; Difference equation simulation.
Runge–Kutta Algorithms; Stability domains of Runge–Kutta Algorithms; extrapolation technique; Solver for Differential Algebraic Equations
Time moments, Markov parameters, Pade Approximation, Routh Approximation, Parametric estimation.
Random variables, Probability distribution function, Probability Density function, Autocorrelation, Cross-correlation, White Noise, Random process, Markov process; Markov chain.
Bayesian statistics; simulated annealing; Monte-Carlo simulation.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	H	H	M	M
CO2	H	H	H	H	H	M
CO3	H	M	L	H	L	L
CO4	H	H	H	H	H	M
CO5	H	H	H	H	H	L

[High: H; Moderately: M; Low: L]

References

1. François E. Cellier and Ernesto Kofman, Continuous Systems Simulation, Springer 2006.
2. Sheldon M. Ross, Simulation, Academic Press, 5th Edition, 2013.
3. Kai Velten, Mathematical Modeling and Simulation, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim, 2009.

23/7/15 Nagi 24/7/15

Course Name:	Wide Area Monitoring Systems	Contact hours	3+1+0
Course Code:	EE21405	Credit of the Course	4
Pre-requisite knowledge	Basic Electrical Engineering.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the needs and fundamental of wide area measurement system (WAMS).
CO2. Application of WAMS for protection and control.
CO3. Communication protocols for WAMS.
CO4. Development of smart meters and phasor measurement units.
CO5. Applied data analytics for PMU data.

Syllabus

Module 1 Fundamentals of WAMS

Introduction to WAMS and Synchrophasor technology, Need and evaluation of WAMS, Basic principles for wide area, local area - measurement, instrumentation, control and protection in real-time, Concept of synchronization and time protocols.

Module 2 Metering aspects for WAMS

Frequency measurement units, phasor measurement units: - time domain, frequency domain and time-frequency domain techniques, phasor data concentrators - hardware and software, IEEE standards for phasor estimation, Smart meters, meter tapering.

Module 3 Communication aspects for WAMS

Elements of communication and networking: architectures, standards and adaptation of power line communication (PLCC), zigbee, GSM, and more; machine to-machine communication models for the smart grid; Home area networks (HAN) and neighborhood area networks (NAN); reliability, redundancy and security aspects, IEEE standards for phasor communication.

Module 4 Application of WAMS

Wide area power system stabilizers; Synchrophasor applications in power system protection and emergency control; Optimal placement of phasor measurement units; State estimation; Use of ML techniques for analysing wide area measurement data, Real-time monitoring and control of power system stability, Case studies of various country WAMS.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	L	L	L
CO2	M	L	L	H	M	H
CO3	H	H	H	L	L	H
CO4	M	H	M	L	H	H
CO5	L	L	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. A. G. Phadke and J. S. Thorp, Synchronized Phasor Measurements and their Applications, Springer, 2008.
2. M. Shadidehpour and Y. Wang, Communication and Control in Electric Power System, Wiley, 2003.
3. P. Kundur, Power System Stability and Control, McGraw-Hill, 1995.
4. P. M. Anderson and A. A. Fouad, Power System Control and Stability, 2nd Edition, Wiley, 2003.
5. Hsiao – Dong Chiang, Direct Methods for Stability Analysis of Electric Power Systems: Theoretical Foundation, BCU Methodologies, and Applications, Wiley, 2011.

Course Name:	Sensors and Interfacing	Contact hours	3+1+0
Course Code:	EE21406	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Control System Design.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Perform the comparative assessment among different sensors using various standards.
CO2. Understand the main characteristics of different sensors with the specifications.
CO3. Understand the concept of interfacing employing sensors and their applications.
CO4. Evaluate and design sensor using IoT methodologies and will be able to apply to real-time systems.
CO5. Perform sensors interfacing tasks with real-time testbeds.

Syllabus

Module 1 Introduction to Sensors

Operating principles, Parameters, Characterization, Transducers and its types, Optical sensors, Actuators, Working, Classifications and Applications.

Module 2 Sensors and Signal Conditioning Circuits

Mechanical and electromechanical, Thermal, Magnetic, Radiation, Smart Sensors.
Signal Conditioning Circuits: Instrumentation amplifier, Precision rectifiers, Filters, Charge amplifier, D/A Converter: Weighted resistor type, R-2R ladder network, A/D converter: flash type, successive approximation type, delta-sigma type, dual slope type, V-to-I and I-to-V converters, Phase-sensitive rectifiers application to rectilinear motion sensors.

Module 3 Applications

Automotive Sensors, Home appliance sensors, Aerospace Sensors, Medical diagnostic sensors, Environmental monitoring.

Module 4 Introduction to Sensor Networks

Background of sensor network technology, Brief historical survey of sensor networks, Challenges, Basic sensor network, Architectural elements, Applications of sensor networks, Hardware components, Energy consumption of sensor nodes, Operating system and execution environment, Examples of sensor nodes, Sensor Coverage and Sensor Fusion.

Module 5 Sensors Interfacing

Sensor network scenarios, Optimization goals, Design principles, Service interfaces, Gateway concepts, IoT-Introduction, Definition and Characteristics, Physical design, Logical design, IoT levels and deployment templates, Raspberry Pi, Intel EDISON, pcDuino, Beagle Bone Black, Cubie board and its interfacing.
8051 microcontroller review, peripheral interfacing to 8051 microcontroller.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	H	M	H	M	H
CO2	H	H	H	M	H	H
CO3	L	M	H	L	M	M
CO4	M	M	M	M	H	M
CO5	H	L	M	L	L	H

[High: H; Moderately: M; Low: L]

References

1. H. Kerl, A. Willig, Protocols and Architectures for Wireless Sensor Network. John Wiley and Sons, 1st Edition, 2007.
2. A. Bahga, V. Madisetti, Internet of Things. 1st Edition 2014.
3. K. Sohraby, D. Minoli, T. Zanti, Wireless Sensor Network: Technology, Protocols and Application, John Wiley and Sons. 1st Edition, 2007.

Course Name:	Data Communication and Networking	Contact hours	3+1+0
Course Code:	EE21407	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Computer programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the fundamentals of data communication, networking, and their importance.
CO2. Assess different services/features of various protocol layers in data networks.
CO3. Demonstrate switched networks and associated communication protocols.
CO4. Analyze TCP/IP and their protocols.
CO5. Demonstrate the principle and operations of protocols used in application layer.

Syllabus

Module 1 Data Communications

Components, Direction of Data flow, Networks, Types of Connections, Topologies, Protocols and Standards – ISO / OSI model, Example Networks such as ATM, Frame Relay, ISDN Physical layer: Transmission modes, Multiplexing, Transmission Media, Switching, Circuit Switched Networks, Datagram Networks, Virtual Circuit Networks.

Module 2 Data Link Layer

Introduction, Framing, and Error Detection and Correction, Parity – LRC – CRC Hamming code, Flow and Error Control, Noiseless Channels, Noisy Channels, HDLC, Point to Point Protocols, 111 Medium Access sub-layer: ALOHA, CSMA/CD, LAN – Ethernet IEEE 802.3, IEEE 802.5 – IEEE 802.11, Random access, Controlled access, Channelization.

Data encoding and transmission: unipolar/polar/bipolar encoding, B8ZS, HDB3, analog-to-digital conversion: PAM & PCM, transmission medium: LAN/OSI, Transmission Impairment, ADTs and Protocol Design.

Data link control: line discipline (ENQ/ACK & Poll/Select), flow control (stop-and-weight/sliding window), and error control (stop-and-wait ARQ/sliding window ARQ).

Module 3 Network Layer

Logical Addressing, Internetworking, Tunneling, Address mapping, ICMP, IGMP, Forwarding, routing algorithms: adaptive, non-adaptive, and hybrid algorithms, Uni-Cast Routing Protocols, Multicast Routing Protocols.

Module 4 Transport Layer

Process to Process Delivery, UDP and TCP protocols, Data Traffic, Congestion, Congestion Control, QoS, Integrated Services, Differentiated Services, QoS in Switched Networks, Multiplexing and demultiplexing in transport layer: TCP (connection oriented) & UDP (connection less).

Module 5 Application Layer

Domain name space, DNS in internet, electronic mail, SMTP, FTP, WWW, HTTP, SNMP.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	M	H	L	L
CO2	M	H	H	H	M	L
CO3	M	L	H	L	M	M
CO4	M	H	H	H	M	L
CO5	H	L	L	M	M	L

[High: H; Moderately: M; Low: L]

References

1. Forouzan, Behrouz A. Data Communications and Networking with TCP/IP Protocol Suite. McGraw-Hill US Higher Ed USE, 2021.
2. A. S. Tanenbaum, Computer Networks, 4th Edition, Pearson Education, PHI, 2018.

Course Name:	Optimal and Robust Control	Contact hours	3+1+0
Course Code:	EE21408	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Control System Design.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand Calculus of Variation and different forms of performance measures.
CO2. Model and analyze the linear regulator/tracking problems.
CO3. Develop state estimator and design of control systems using the same.
CO4. Perform robustness study of uncertain multivariable systems.
CO5. Design robust control framework using computer-aided tools and comments of stability.

Syllabus

Optimal Control Systems and Performance Indices, Optimal Control of linear systems with Quadratic Performance Index, Optimal State Regulator Design through Matrix Ricatti Equation, Properties and application of the Optimal Regulator, Linear Quadratic Gaussian(LQG) Control, State Estimator Design and application, Frequency Shaping.

Robust Control Systems and System sensitivity, Uncertainty and Robustness, Structured and Unstructured uncertainties, InternalStability.

Kharitnov's method, Stability robustness and Performance robustness of Control Systems, μ-Synthesis, H_2/H_∞ Control/Loop Shaping, Linear Matrix Inequalities (LMI).

Variable structure control (VSC), Sliding mode control (SMC), Fast Terminal Sliding Mode control, Active Disturbance Rejection Control (ADRC), Case studies.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	L	L	L
CO2	M	H	M	M	L	L
CO3	L	M	H	M	L	L
CO4	L	L	M	H	H	M
CO5	L	L	M	M	H	H

[High: H; Moderately: M; Low: L]

References

1. D. E. Kirk, Optimal Control Theory: An Introduction, Prentice-Hall, 2004.
2. B.D.O. Anderson and J.B. Moore, Optimal Control: Linear Quadratic Methods, Dover Publications, 2014.
3. F.L. Lewis, D. Vrabie and V.L. Syrmos, Optimal Control, 3rd edition, Wiley & Sons, 2012.
4. Kemin Zhou and John C. Doyle, Essentials of Robust control, Prentice Hall, 1999.
5. K. Zhou, J. C. Doyle and K. Glover, "Robust and Optimal Control", Prentice-Hall, Englewood Cliffs, NJ, 1995.

23/7/25 Rnegi 24/7/25

Course Name:	Optimization Techniques	Contact hours	3+1+0
Course Code:	EE21301	Credit of the Course	4
Pre-requisite knowledge	Mathematics.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Introduce basic optimisation techniques.
CO2. Understand the linear optimisation techniques and their applications.
CO3. Understand the non-linear optimisation techniques and their applications.
CO4. Study evolutionary based algorithms for optimisation techniques.

Syllabus

Classical optimization techniques: Single variable optimization, multivariable optimization with constraints and without constraints, necessary and sufficient conditions.

Linear programming (LP): Two variable problems-graphical solutions, formulation of LP problems in more than two variables, standard form, Simplex algorithm, special cases-2 phase's method, Big-M method, duality and dual LP problems. Application of LP in Transportation problem-balanced and unbalanced transportation problems. Use of North West corner rule, least cost method, Vogel approximation method. Assignment problems- Hungarian method.

Non-linear programming (NLP): Philosophy of numerical methods, search methods for one dimensional problems- Fibonacci and Golden section methods. Unconstrained and constrained optimization, univariate method, Pattern search method, Steepest descent method, cutting plane method, penalty function method, basic idea of dynamic programming.

Evolutionary algorithms (EA): Genetic algorithm, particle swarm optimization, Tabu search, simulated annealing and ant colony optimization, Multi objective optimization using EA, Pareto solutions.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	L	L	H	M
CO2	H	L	L	L	M	H
CO3	H	M	M	M	H	H
CO4	H	H	M	M	H	H

[High: H; Moderately: M; Low: L]

References

1. S.S. Rao, Engineering Optimization: Theory and Practice. New York: Wiley. 2009.
2. K. Deb, Multiobjective Optimization using Evolutionary Algorithms. New York; Wiley. 2002.
3. G.P. Liu, J.B. Yang and J.F. Whidborne, Multiobjective Optimization and Control. PHI. 2008.
4. A. D. Belegundu, and T. R. Chandrupatla, Optimization Concepts and Applications in Engineering, Pearson Education (Singapore). 2003.
5. R. L. Rardin, Optimization in Operation Research. Prentice-Hall. 1999.
6. A. Schirisieer, Theory of linear and integer programming, John Wiley and Sons, 1998.
7. David G. Luenberger, Yinyu Ye, Linear and Nonlinear programming, Springer, 2008.

Course Name:	Digital Signal Processing	Contact hours	3+1+0
Course Code:	EE21409	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Signals and Systems.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Describe digital signals/systems and their significance.
CO2. Understand and apply Z-transform tool for analysis.
CO3. Perform signal processing using different digital transform methods, e.g., DFT, FFT.
CO4. Develop digital filters and interpret shortcomings in real-time implementation.
CO5. Design optimal filter to perform the signal processing tasks.

Syllabus

Introduction to Digital Signal Processing, Digital Systems and its characterization, Impulse Response, Convolution, Inverse System, Stability, FIR & IIR, Recursive and Non-recursive, Introduction to multi-rate signal processing.

Z-transform revisited, Discrete-time Fourier Transform (DTFT), Discrete Fourier Transform (DFT), Fast Fourier Transform, Discrete wavelet transform andapplications, Short-time Fourier transform.

Linear phase filters, Complementary transfer function, Analog filter design, IIR Realization, Bi-linear transformation, FIR Realization: Windows techniques, Finite word length effects in FIR and IIR digital filters.

Optimum signals estimation: Sum of squared errors (SSE), mean squared error (MSE), linear mean square error estimation, Optimum FIR filters design and linear prediction.

Recursive-Least-Squares (RLS) Adaptive Filters, RLS FIR Filter Coefficients, Frequency Domain Characteristics of an RLS Narrow-Band Suppression Filter,

Wiener Filtering: concept, random process and properties, Two-Sided (Unconstrained) Wiener Filters, Causal (Shannon-Bode) Wiener Filters, Optimal FIR Wiener Filters.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	L	M	L	L
CO2	L	M	L	M	M	L
CO3	M	H	H	M	H	H
CO4	M	H	M	M	H	M
CO5	L	L	M	M	H	M

[High: H, Moderately: M, Low: L]

References

1. Oppenheim A V and Sehafer R W, "Discrete Time Signal Processing", 2nd Edition, Prentice Hall, 1998.
2. Proakis J G and Manolakis D G, "Digital Signal Processing: Principles, Algorithms, and Applications", 3rd Edition, Pearson Education India, 1996.
3. Oppenheim A V, WillskyA S and Young I T, "Signal & Systems", 2nd Edition, Prentice Hall, 1996.
4. Ifeachor and Jervis, "Digital Signal Processing: A practical approach", Pearson Education India, 1993.
5. DeFatta D J, Lucas J G and Hodgkiss W S, "Digital Signal Processing: A System Design Approach", J Wiley and Sons, Singapore, 2009.

Course Name:	Linear Algebra	Contact hours	3+1+0
Course Code:	EE21320	Credit of the Course	4
Pre-requisite knowledge	Mathematics.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand and explain the algebraic structures.
CO2. Understand and explain the linear transformations.
CO3. Understand theory of vector spaces, sub-spaces, and their properties.
CO4. Understand and explain the various transforms.

Syllabus

Module 1 Algebraic Structures

Sets, functions, Group, homomorphism of groups, Ring, Field, Vector Space, Subspaces, direct sum, metric space, inner product space, L_p space, Banach Space, Hilbert Space. Linear independence, basis, dimension, orthonormal basis, finite dimensional vector spaces, isomorphic vector spaces, Examples of finite and infinite dimensional vector spaces, RN, CN.

Module 2 Linear Transformations

Linear Transformations, four fundamental subspaces of linear transformation, inverse transformation, rank nullity theorem, Matrix representation of linear transformation, square matrices, unitary matrices, Inverse of a square matrix, Change of basis, coordinate transformation, system of liner equations, existence and uniqueness of solutions, projection, least square solution, pseudo inverse.

Module 3 Matrix Methods and Transforms

Eigen values, Eigen vectors, Generalized Eigen vectors, Diagonalizability, orthogonal diagonalization, Symmetric, Hermitian and Unitary matrices (transformations), Jordan canonical form, Fourier basis, DFT as a linear transformation, Translation invariant linear transformation, wavelet basis, wavelet transforms.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	L	H	H	M
CO2	H	M	L	M	M	L
CO3	M	H	H	H	M	L
CO4	M	H	L	H	H	M

[High: H; Moderately: M; Low: L]

References

1. G. F. Simmons, Introduction to Topology and Modern Analysis, McGraw Hill, 1983 (Reprint Edition).
2. Frazier, Michael W. An Introduction to Wavelets through Linear Algebra, Springe, 1999.
3. Hoffman Kenneth and Kunze Ray, Linear Algebra, Pearson, 2015.

23/7/25 24/7/25

Course Name:	Mechatronics and Automation	Contact hours	3+1+0
Course Code:	EE21410	Credit of the Course	4
Pre-requisite knowledge	Electrical Measurements, Control System Design, Robot Control.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand and integrate mechatronic components to meet product requirements.
CO2. Develop mechatronic products.
CO3. Understand different control actions and their applications in industrial automation.
CO4. Demonstrate the application of SCADA in industrial automation problems.
CO5. Perform the analysis of automation system augmented with Hardware-in-Loop (HIL).

Syllabus

Introduction: the design process, Measurement systems, Control systems, Programmable logic controller, Example of mechatronic systems, Mathematical models: Mechanical, Electrical, Fluid, and Thermal Systems, Rotational and translational, electro-mechanical, pneumatic-mechanical, hydraulic-mechanical, micro electro mechanical system, Dynamic responses of system: first order, second order system, Performance measures.

Introduction to programmable logic controller (PLC), Principles of operation, Programmable logic controller Architecture and specifications, Programmable logic controller hardware components, Programmable logic controller ladder diagram, PLC programming: Simple instructions, Manually operated switches, Mechanically operated switches, Latching relays, Applications of PLC.

Industrial automation: transition from Industry 1.0 to Industry 5.0, Distributed computing and SCADA, Hardware-in-the-loop (HIL).

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	H	M	L	L
CO2	M	M	H	H	L	L
CO3	M	L	L	M	M	L
CO4	H	H	M	H	M	H
CO5	H	H	L	L	L	H

[High: H; Moderately: M; Low: L]

References

1. Gary Dunning, Introduction to Programmable Logic Controllers, 3rd Edition, Cengage Learning, 2007.
2. J. Webb, Programmable Logic Controllers: Principles and Applications, 5th Edition, Prentice Hall of India, 2012.
3. W. Bolton, Mechatronics: electronic control systems in mechanical and electrical engineering, Longman, Singapore, 1999.
4. A. Hamdan, A. Harraf, A. Buallay, P. Arora, and H. Alsabatin, From Industry 4.0 to Industry 5.0: Mapping the Transitions, Studies in Systems, Decision and Control (SSDC, volume 470), 2023.

Semester-II

Course Name:	Cyber Security of Physical Systems	Contact hours	3+0+2
Course Code:	EE22131	Credit of the Course	4
Pre-requisite knowledge	Introduction to Cyber-physical system, Computer programming skills, Control System Design, Linear Algebra.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Describe the modelling of CPS in state-space.
CO2. Define and model different possible cyber-attacks in the CPS.
CO3. Apply detection strategies for cyber-attacks.
CO4. Securely analyse the vulnerability of CPS against cyber-attacks.
CO5. Increase cyber-tolerance of CPS employing resilient control framework.

Module 1 Cyber-attacks and modelling

Smart systems, SQL injection, OS Command, Cyber-attacks: deception type, DoS attack, false-data injection, input delay switching (TDS) attack, Sensors and actuator attacks, Unified modelling of CPS attack; Case of undetectable attacks, Security issues of Industrial Cyber-physical Systems, Threat model and its effect on control scheme, Packet-data loss.

Module 2 Attacks detection and estimation

Attacks detection strategies, Model-based attacks detection, Estimation of detectable attacks. Design of observers under sensor and actuator attacks, Design of observer for distributed environment under different attacks.

Module 3 Attacks mitigation

Mitigation control approaches, Centralized and distributed control, advanced control techniques, event-based mitigation approach, Fixed-time cyber-tolerant control, Fault-tolerant control. Security of MAS: switched-based modelling, Actuator/Sensors attacks and mitigation.

Case studies: Smart Grid, Swarms of UAVs, Mobile robots, System Operators, Biological testbeds, etc.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	M	L	L	M
CO2	H	H	M	M	M	L
CO3	H	H	H	H	H	M
CO4	H	H	H	H	H	M
CO5	H	H	H	H	H	H

[High: H; Moderately: M; Low: L]

References

1. F. Pasqualetti, F. Dörfler and F. Bullo, "Attack Detection and Identification in Cyber-Physical Systems," IEEE Transactions on Automatic Control, vol. 58, no. 11, pp. 2715-2729, Nov. 2013.
2. H. Fawzi, P. Tabuada and S. Diggavi, "Secure Estimation and Control for Cyber-Physical Systems Under Adversarial Attacks," IEEE Transactions on Automatic Control, vol. 59, no. 6, pp. 1454-1467, June 2014.
3. Yilin Mo, Rohan Chabukswar and Bruno Sinopoli "Detecting Integrity Attacks on SCADA Systems" IEEE Transactions on Control System Technology, Vol. 22, No. 4, 2014.
4. F. Pasqualetti, F. Dörfler and F. Bullo "Control Theoretic methods for Cyber Physical Security", IEEE Control System Magazine, pp. 110-127, Feb. 2015.
5. Y. Maa, Z. Li, X. Xie, D. Yue. Adaptive consensus of uncertain switched nonlinear multi-agent systems under sensor deception attacks. Chaos, Solitons and Fractals, vol. 175, 113936, 2023.
6. Technical Reports (real-time data) of -- for study and analysis.
7. Journals/Conference Proceedings.

24/7/25
22/7/25

Cyber Security of Physical Systems Laboratory

List of Experiments

1) Development of cyber physical system to study the impacts of different attacks.
2) Study of CPS performance against
   a) Sensor deception attacks
   b) DoS attack
   c) Actuator attack
3) Development of resilient mitigation control framework employing
   a) Artificial Intelligence
   b) Machine learning algorithms.
4) Development of CPS-based multi-agent framework (leader-follower formation).

Course Name:	Internet-of-Things (IoT)	Contact hours	3+0+2
Course Code:	EE22132	Credit of the Course	4
Pre-requisite knowledge	Linear Algebra, Computer programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand different terminologies and architecture of IoT Systems.
CO2. Utilize sensors and actuators in IoT design.
CO3. Apply different communication protocols in IoT design.
CO4. Apply skills in solving industrial problems.
CO5. Determine the suitable IoT devices based on the real-time case studies.

Syllabus

Module 1 Introduction to IoT and architecture

Fundamentals of IoT: Introduction, Sensing, Actuation, Basics of Networking, Definitions & Characteristics of IoT, IoT Architectures, Physical & Logical Design of IoT, Enabling Technologies in IoT, History of IoT, About Things in IoT, The Identifiers in IoT, About the Internet in IoT, IoT frameworks, IoT and M2M communication.

Module 2 Python programming and implementation

Introduction to Python Programming, Introduction to Raspberry Pi, Interfacing Raspberry Pi with Basic Peripherals, Implementation of IoT with Raspberry Pi.

Module 3 Communication protocols of IoT

Wireless Technologies for IoT: WPAN Technologies for IoT: IEEE 802.15.4, Zigbee, HART, NFC, Z-Wave, BLE, Bacnet, Modbus.

IP Based Protocols for IoT IPv6, 6LowPAN, RPL, REST, AMPQ, CoAP, MQTT. Edge connectivity and protocols.

Module 4 Applications of IoT

Applications of IoT: Internet of Medical Things (IoMT), Industrial Internet of Things (IioT), Power generation, transmission, digital twin, aircraft, etc.

Mapping between COs and Pos

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	L	H	L	M	L
CO2	H	H	M	L	M	L
CO3	H	H	H	L	M	L
CO4	L	H	H	M	H	H
CO5	M	H	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Technologies, Platforms, and Use Cases", CRC Press, 2017.
2. Hakima Chaouchi, "The Internet of Things Connecting Objects to the Web" ISBN: 978-1-84821-140-7, Wiley, 2010.
3. Olivier Hersent, David Boswarthick, and Omar Elloumi, "The Internet of Things: Key Applications and Protocols", Wiley, 2012.
4. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on-Approach)", 1 st Edition, VPT, 2014.
5. J. Biron and J. Follett, "Foundational Elements of an IoT Solution", O'Reilly Media, 2016.
6. Keysight Technologies, "The Internet of Things: Enabling Technologies and Solutions for Design and Test", Application Note, 2016.

Internet-of-Things (IoT) Laboratory

List of Experiments

1) Universal embedded microcontroller development board.
2) 8051 Daughter board and ARM 7 Daughter board
3) Traffic Light Controller Interface
4) Accelerometer Sensor Interface
5) Humidity + IR Sensor Interface
6) Raspberry Pi 3B+ Original

23/7/25 24/7/25

Course Name:	Systems Identification and Estimation	Contact hours	3+1+0
Course Code:	EE22313	Credit of the Course	4
Pre-requisite knowledge	Linear algebra, Computer programming skills, Control System Design.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Apply basics of probability theory in system identification and system state estimation.
CO2. Methods for non-parametric system identification and model validation.
CO3. Estimation of parametric models and validation of the obtained system models.
CO4. Design state estimator for linear deterministic and stochastic systems.
CO5. Apply Kalman filter in various variants for system state and parameter estimations.

Syllabus

System Identification - Motivation and Overview, random variables and stochastic processes, stochastic static models, Disturbance models - random processes, representation of stationary processes, white-noise process, auto-covariance function (ACF), ARMA models, Parametric model structures - ARX, ARMAX, etc.

The identification problem, classical methods of identification of transfer functions models, Linear Regression - Least Squares estimates, Statistical properties of LS Estimates, Weighted Least Squares, Recursive Least Squares, Maximum Likelihood Estimation, Minimum variance algorithm, stochastic approximation.

Estimation of non-parametric models - impulse, step, response coefficients, frequency response models, Estimation of parametric models - notions of prediction and simulation, predictors for parametric models, prediction-error methods, Instrumental Variable method.

Kalman filter (KF) for state estimation, Proof of optimality of KF, Moving Horizon Estimation (MHE), Extended Kalman filtering (EKF), Unscented Kalman filter (UKF), and Cubature Kalman filter (CKF), EKF, UKF, CKF, and MHE Comparison.

Gauss Markov, Model for vector random processes. Model Structure Selection and Diagnostics - estimation of delay and order, residual checks, properties of parameter estimates, model validation.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	H	H	H	L
CO2	H	H	M	H	H	L
CO3	H	H	M	H	H	L
CO4	H	H	M	H	H	L
CO5	H	H	H	H	H	L

[High: H; Moderately: M; Low: L]

References

1. Pieter Eykhoff, "Trends and progress in system identification," Pergamon, 1981.
2. Raman K. Mehra, "System identification advanced and case studies," Elsevier Science; edition, 1976.
3. Jason L. Speyer, Walter H. Chung, "Stochastic processes, estimation and control," Society for Industrial and Applied Mathematics, 2008.
4. Gregory F. Lawler, "Introduction to stochastic processes," Chapman' and Hall/CRC; 2 edition, 2006.

Course Name:	Adaptive Control	Contact hours	3+1+0
Course Code:	EE22314	Credit of the Course	4
Pre-requisite knowledge	Control System Design, Optimal and Robust Control.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Explain self-tuning, adaptive, and robust adaptive control systems.
CO2. Application of robust adaptive control systems.
CO3. Understand adaptive control systems, their development and properties.
CO4. Understand methods and tools for stability analysis of adaptive and learning systems.

Syllabus

Introduction: Linear feedback, effects of process variations, adaptive schemes, adaptive control problem.
Real-Time Parameter Estimation: Least squares and regression models, estimating parameters in dynamical systems, simulation of recursive estimation.
Self-tuning Regulators (STR): Pole placement design, indirect STR, direct STR, stochastic and predictive STR, applications.
Model-reference Adaptive Control (MRAC): The MIT rule, determination of adaptive gain, design of MRAS' using Lyapunov theory, BIBO stability, output feedback, relation between MRAS and STR, applications.
Gain Scheduling: The principles, design of gain scheduling controllers, nonlinear transformations, and applications.
Robust adaptive control scheme, averaging-based analysis, adaptive control of nonlinear systems, Practical issues and implementation, commercial products and applications, Case studies.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	L	L	L	H	H	M
CO2	M	H	H	H	H	M
CO3	L	L	L	H	H	M
CO4	L	L	M	H	H	M

[High: H; Moderately: M; Low: L]

References

1. K. J. Astrom, B. Wittenmark, "Adaptive Control, Prentice Hail," 2 edition, 1994.
2. P. A. Ioannou, J. Sun, "Robust Adaptive Control, Dover Publication," 2 edition, 2012.
3. H. K. Khalil, "Nonlinear Systems," Prentice Hall; 3 edition, 2001.
4. K. S. Narandra, A. M. Annaswamy, "Stable Adaptive Systems," Prentice Hall, EnglewoodClis, NJ, 2005.
5. S. Sastry, M. Bodson, "Adaptive Control: Stability, Convergence and Robustness," Dover Publications, 2011.

Course Name:	Model Predictive Control (MPC)	Contact hours	3+1+0
Course Code:	EE22401	Credit of the Course	4
Pre-requisite knowledge	Mathematics, Control System Design, Digital Control.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Explore the features of Model Predictive Control (MPC).
CO2. Demonstrate the dynamic behaviour of plant having MPC integrated.
CO3. Develop state estimator and perform linear quadratic theory.
CO4. Understand the prediction time horizon and reducing the same.
CO5. Analyze the control engineering problems using MPC.

Syllabus

Review of Models and Modelling, Single input single output (SISO) control; model based control; multivariable control strategies, model forms for model predictive control, Review of Optimal Control theory.

Parametric models of dynamical systems, Least squares and regression models, Parametric Identification, Adaptive control problem.

Model stability; Observability and controllability, Representing uncertainty, White, Colored and integrating noise.

Dynamic matrix Control: Future prediction, Objectives and Constraints, Step response of MPC, Closed-loop properties of model predictive control.

Linear quadratic problem, Dynamic Programming Solution: Min-Max Optimal Control, Infinite Horizon LQ Problem, Predictive control strategy, prediction model, constraint handling prediction equations, Stability, Reducing horizon MPC for constraint linear systems.

Unconstrained Linear Quadratic Regulator, Linear Systems with Control and State Constraints, Constrained Nonlinear time-invariant and time-Varying systems, Sub-optimal MPC.

State-space MPC; deterministic formulation; State Feedback control, Separation principle; State-space model of output feedback MPC, Implementation of output feedback MPC.

Review of stochastic process: stationary & non-stationary, Optimal State Estimation of linear systems with Kalman filtering (KF) and its variants, Case studies.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	H	H	H	L
CO2	M	L	L	H	H	M
CO3	H	H	H	H	M	L
CO4	H	H	H	H	H	L
CO5	H	H	H	M	H	L

[High: H; Moderately: M; Low: L]

References

1. J.B. Rawlings, D.Q. Mayne and M.M. Diehl, Model Predictive Control: Theory, Computation, and Design, Nobb Hill, 2018.
2. E.F. Camacho and C. Bordons, Model Predictive Control, Springer, 2007.
3. J.A. Rossiter, Predictive Control: a practical approach", CRC Press, 2003.

23/12/25 24/7/25

Course Name:	Nonlinear Control	Contact hours	3+1+0
Course Code:	EE22402	Credit of the Course	4
Pre-requisite knowledge	Signals and Systems, Network theory, Control System Design, Mathematics.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Understand nonlinear control systems with its distinct features.
CO2. Formulate the mathematical model of nonlinear systems.
CO3. Perform stability analysis of nonlinear control systems with various approaches.
CO4. Develop active disturbance rejection controller for better shaping of system dynamics.
CO5. Design/formulate resilient control law to correlate between design parameters and the system performance.

Syllabus

Module 1 Introduction

Affine System, Time-varying affine system, Features of nonlinear system, Limit cycle, Equilibrium point, Existence of periodic orbits: Poincare-Bendixson criteria, Phase portrait, Switched-mode nonlinear system, Positive real systems, Norms of signal, State-space modelling.

Module 2 Stability

Asymptotic stability, Exponential Stability, Linearization and Local stability of nonlinear systems, Global Stability, Ultimate Uniform Stability, Input to State Stability, Finite-time stability, Lyapunov Direct method, LaSalle Invariant Set theory, Barbalet Corollary, Kalman-Yakubovich-Popov Lemma, Stability of perturbed nonlinear systems, Input-output stability, Small-gain theorem, Passivity.

Module 3 Describing Function

Inherent and Intentional nonlinearities, Describing function and Descriptor system, Describing function modelling of pure relay, dead-zone, relay with dead-zone, saturation, dead-zone with saturation, and Backlash nonlinearities, Extension of Nyquist Stability Criteria to nonlinear systems, findings of existence of limit cycle and calculation of periodic orbits, Stability analysis.

Module 4 Disturbance Observer

Matched/unmatched disturbances, Luenberger Disturbance Observer, Nonlinear Disturbance Observer, Asymptotic Stability of Disturbance Observer: Lyapunov approach.

Module 5 Nonlinear Control

Feedback linearization and the canonical form, H-infinity Control, Back stepping method for non-feedback linearizable systems, Control of physical systems, Simulation studies.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6
CO1	H	H	M	H	M	L
CO2	H	L	L	M	M	H
CO3	H	H	M	H	H	L
CO4	H	H	H	H	H	L
CO5	H	H	H	H	M	L

[High: H; Moderately: M; Low: L]

References

1. H. K. Khalil, Nonlinear systems, Prentice Hall, 3rd Edition, 2002.
2. Shankar Sastry, Nonlinear Systems; Analysis, Stability and Control, Springer, 1999.
3. M. Vidyasagar, Nonlinear systems analysis, 2nd Edition, Society of Industrial and Applied Mathematics, 2002.
4. Slotine and Li, Applied Nonlinear Control, Prentice Hall, 1991

5. S. Li, J. Yang, W.H. Chen, and X. Chen, Disturbance Observer-Based Control: Methods and Applications, CRC Press, 2014.
6. W. H. Chen, J. Yang, L. Guo and S. Li, "Disturbance-Observer-Based Control and Related Methods -- An Overview," in IEEE Transactions on Industrial Electronics, vol. 63, no. 2, pp. 1083-1095, Feb. 2016.
7. H. Sira-Ramírez, A. Luviano-Juárez, M. Ramírez-Neria, and E. W. Zurita-Bustamante, Active Disturbance Rejection Control of Dynamic Systems, Elsevier, 2017.

Course Name:	Network Security	Contact hours	3+1+0
Course Code:	EE22403	Credit of the Course	4
Pre-requisite knowledge	Computer programming skills.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Understand Network Security Concept and its need thereof in Communication/data transmission.
CO2. Identify the security issues in the network and able to design solutions.
CO3. Classify different malicious attacks in data transmission medium.
CO4. Understand Various Encryption mechanisms for secure data transmission.
CO5. Perform different web security tasks.

Syllabus

Need of security, security attacks, Services and Mechanisms, Network Security, Models.
Symmetric Ciphers, Substitution &Transposition Techniques, Block Cipher, DES, Triple DES
Public Key Cryptography - Need and Principles of Public Key Cryptosystems, RSA Algorithm, Key Distribution and Management, Diffie-Hellman Key Exchange, Digital Signatures.
Authentication Requirements, Message Authentication Codes, Hashes, MD5 & SHA.
User Authentication: Password, Certificate based & Biometric Authentication, Kerberos.
Network Security – Firewalls, IP Security, VPN 4, Intrusion Detection, Web Security, SSL, TLS.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6
CO1	M	L	L	L	L	L
CO2	M	H	M	L	H	L
CO3	M	H	M	L	L	L
CO4	M	M	M	L	L	M
CO5	H	H	M	L	L	M

[High: H; Moderately: M; Low: L]

References

1. William Stalling, "Cryptography & Network Security", 7th Edition, Pearson, 2017.
2. Atul Kahate, "Cryptography & Network Security", McGraw Hill, 2017.
3. Behrouz A. Forouzan, "Introduction to Cryptography & Network Security", McGraw Hill, 2008.

Course Name:	Power Systems Dynamics, Stability and Control	Contact hours	3+1+0
Course Code:	EE22404	Credit of the Course	4
Pre-requisite knowledge	Basic Electrical Engineering, Optimization techniques.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the dynamical modelling and stability notions of power system.
CO2. Analyze the impact of real-time data on power system stability assessment.
CO3. Explore advanced stability modelling and control techniques with real-time data.
CO4. Investigate case studies and real-world applications of real-time data in enhancing power system stability.
CO5. Develop critical thinking and problem-solving skills in the field of power system stability.

Syllabus

Module 1 Introduction

Review of power system stability concepts, Transient, steady-state, and dynamic stability, Factors affecting power system stability, Basics of data acquisition and communication: SCADA and WAMS, Concept of synchronization and time protocols.

Module 2 Stability assessment with WAMS

Detailed models for synchronous generators and power system components, Incorporating WAMS synchronized data into stability models, Real-time stability assessment using synchrophasor data, State estimation & situational awareness, WAMS applications for early warning and preventive control.

Module 3 Wide area control techniques

Adaptive control techniques with WAMS, wide area control using synchrophasor data, coordinated control schemes for stability enhancement, challenges faced in implementing WAMS for stability.

Module 4 Cybersecurity and Resilience of WAMS

Cyber threats to WAMS and power grids, security measures, best practices, and resilience planning, Case study: Discussion or simulation of a practical case study involving stability analysis of a power system enabled with WAMS technology.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	**H**	**H**	**L**	**M**	**L**	**L**
CO2	M	M	M	L	L	M
CO3	H	H	M	L	M	M
CO4	H	M	M	L	L	M
CO5	M	M	L	M	H	M

[High: H; Moderately: M; Low: L]

References

1. H. Bevrani, M. Watanabe, and Y. Mitani, Power System Monitoring and Control, Wiley, 2014.
2. M. Shadidehpour and Y. Wang, Communication and Control in Electric Power System, Wiley, 2003.
3. P. Kundur, Power System Stability and Control, McGraw-Hill, 1995.
4. P. M. Anderson and A. A. Fouad, Power System Control and Stability, 2nd Edition, Wiley, 2003.

Course Name:	Wireless Communication	Contact hours	3+1+0
Course Code:	EE22405	Credit of the Course	4
Pre-requisite knowledge	Digital Signal Processing, Data Communication and Networking.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Understand the fundamental principles of wireless communication network.
CO2. Demonstrate the evolution of mobile communication technologies.
CO3. Perform comparative study on different topologies used in wireless communication network.
CO4. Demonstrate architecture of different wireless communication networks.
CO5. Understand the emerging trends of wireless networks.

Syllabus

Overview of current wireless systems and standards, cellular communication, different generations and standards in cellular communication system, satellite communication including GPS, wireless local loop, RFID.

Multicarrier modulation, OFDM, MIMO system, diversitymultiplexing trade-off, MIMO-OFDM system, smart-antenna; beamforming and MIMO, cognitive radio, software defined radio, communication relays, spectrum sharing.

Multiuser wireless communications: multiple access: FDMA, TDMA, CDMA, SDMA, CSMA, interference management: power control; multiuser diversity, multiuser MIMO systems.

Wireless personal area networks (Bluetooth, UWB and ZigBee), wireless local area networks (IEEE 802.11, network architecture, medium access methods, WLAN standards), wireless metropolitan area networks (WiMAX).

Mapping between COs and POs

Course outcomes (COs)	Programme outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	H	M	L	L
CO2	M	M	L	L	L	L
CO3	M	M	H	H	L	L
CO4	H	H	M	H	M	H
CO5	M	H	L	L	M	M

[High: H; Moderately: M; Low: L]

References

1. A. J. Goldsmith, Wireless Communications, Cambridge University Press, 2005.
2. D. Tse and P. Viswanath, Fundamentals of Wireless Communications, Cambridge University Press, 2005.
3. A. Molisch, Wireless Communications, John Wiley & Sons, 2005.
4. S. Haykin and M. Moher, Modern Wireless Communications, Pearson Education, 2005.
5. T. S. Rappaport, Wireless Communications, Prentice Hall, 1996.
6. G. L. Stuber, Principles of Mobile Communications, Kluwer, 1996.
7. T. Cover and J. Thomas, Elements of Information Theory, Second Esition, John Wiley & Sons, 2006.

Course Name:	Embedded Systems - Design and Programming	Contact hours	3+1+0
Course Code:	EE22406	Credit of the Course	4
Pre-requisite knowledge	Computer programming skills.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand different processors and memory architecture.
CO2. Design prototype embedded system in real-time.
CO3. Describe architecture of different multi-tasking.
CO4. Demonstrate real-time operating systems and task scheduling with RTOS.
CO5. Establish the communication interfaces on PC-to-PC, PC-to-DSP, and DSP-to-DSP.

Syllabus

Fundamentals of Embedded Systems, CPUs vs. MCUs vs. Embedded Systems, Features of Embedded Systems, Building Embedded Systems using MCUs, Introduction to the mbedTM Platform, mbed Software Development Kit (SDK), Hardware Development Kit (HDK), mbed Development Tools, mbed Worldwide Developer, Comm Moduley, Freedom KL25Z, NXP LPC1768 Hardware Platform, Nordic nRF51822.

ARM Architectures and Processors, ARM Processor Families, ARM Cortex-M Series, Cortex-M0+ Processor, ARM Processor vs. ARM Architectures,ARM Cortex-M0+ Processor, Cortex-M0+ Block Diagram, Cortex-M0+ Memory Map,Bit-band Operations, ARM Cortex-M0+ Processor Instruction Set, ARM and ThumbInstruction Set, Cortex-M0+ Interrupts, NVIC, PortModule and External Interrupts, Cortex Microcontroller Software Interface Standard (CMSIS), Benefits of CMSIS, CMSIS Functions, C Language vs. Assembly Language, Program-Generation Flow, Program Image, Program Data, Data Types, Accessing Data using C and Assembly, Mixed Assembly and C Programming, Embedded Assembly.

Digital Input and Output, Voltages and Logic Values, GPIO Controller, UsingPointer to Access, GPIO, Define Data Structure for Peripherals, Digital IO Examples, UsingLED, Using 7-Segment Display, Using Infrared Emitter/ Detector, Analog Input, Digital-to-Analog Converter, Analog Output, Analog-to-Digital Converter, ADC range, Resolution and Quantization, Sampling Frequency, Input/ Output Analog Signals using mbed, mbed Analog Input, mbed Analog Output.

Timer and Pulse-Width Modulation, Components of a Standard Timer, Compare Mode, Capture Mode, Pulse-Width Modulation Mode, mbed Timer PWM, Serial Communication Overview (SPI, I2C, SPI, CAN), UART Communication, Operating System Overview, Types and Services of Operating Systems, Real-Time Operating System, RTOS Overview, RTOS Task Scheduling, Keil RTX RTOS, RTOS on mbed platform, mbed RTOS API, Using mbed RTOS API for your Project, Thread, Mutex and Semaphore, Communication interface on PC-to-PC, PC-to-DSP, and DSP-to-DSP.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	M	L	L	L
CO2	L	H	H	M	L	L
CO3	M	M	H	M	L	L
CO4	M	M	L	L	H	L
CO5	H	H	M	L	L	M

[High: H; Moderately: M; Low: L]

References

1. Wayne Wolf, Computers as components: Principles of Embedded Computing System Desig, The Morgan Kaufmann Series in Computer Architecture and Design, 2013.
2. Marilyn Wolf, Computers as Components – Principles of Embedded Computing System Design, Third Edition Morgan Kaufmann Publisher (An imprint from Elsevier), 2012.

3. K.J. Ayala, The 8051 Microcontroller: Architecture, Programming, and Applications, Penram Intl, 1996.
4. David Simon, An Embedded Software Primer, Addison Wesley, 2000.

Semester-III

Course Name:	Multi-Agent Systems	Contact hours	3+1+0
Course Code:	EE23401	Credit of the Course	4
Pre-requisite knowledge	Linear Algebra, Control System Design, Digital Control.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Familiarize with modelling, design and building multi-agent systems.
CO2. Understand game theory concepts relevant to multi-agent systems.
CO3. Analyse strategic and social interaction in distributed systems.
CO4. Apply skills in developing control framework for multi-agent systems.
CO5. Perform the study of MAS under adversaries.

Syllabus

Multi-agent systems: Autonomous agents, homogeneous & heterogeneous agents, cooperative vs. competitive agents, intelligent agents, agents and expert systems.

Self-organising systems: Swarms, cellular automata, self-organisation, dimensions of self-organisation, parameter and policy modifications.

Graph representation of MAS, modelling of multi-agent CPS, Fault tolerance and resilient strategies, Multi-agent encounters, Dominant Strategy, Nash Equilibrium Strategy, Pareto Optimal Strategy, The Prisoners' Dilemma, dependence relations in multi-agent systems

Self-Organisation Mechanisms: Flocking behaviours, Reinforcement learning, cooperation, social and economic approaches, fully artificial mechanisms.

Social interaction: Distributive, redistributive, and procedural justice, socio-cognitive trust framework, opinion formation, social networks (ring, all-to-all, random graphs, small-world, and scale-free networks), Applications of MAS.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	H	L	M	H	M
CO2	M	H	M	M	H	L
CO3	M	L	M	L	M	H
CO4	M	L	L	H	H	H
CO5	H	M	M	H	H	H

[High: H; Moderately: M; Low: L]

References

1. Wooldridge, Michael. An introduction to multi-agent systems. John wiley & sons, 2009.
2. Pitt, Jeremy. Self-organising multi-agent systems: Algorithmic foundations of cyber-anarcho-socialism. World Scientific, 2021.
3. Magdi S. Mahmoud, Multiagent Systems: Introduction and Coordination Control, 1st Edition, CRC Press, 2020.
4. Frank L. Lewis, Hongwei Zhang, Kristian Hengster-Movric, Abhijit Das, Cooperative Control of Multi-Agent Systems: Optimal and Adaptive Design Approaches, Springer, 2014.

Course Name:	Quantum Computing	Contact hours	3+1+0
Course Code:	EE23402	Credit of the Course	4
Pre-requisite knowledge	Computer programming skills, Linear Algebra.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Understand the physics/mechanics of quantum computing.
CO2. Perform comparative assessment between classical and quantum computing theories.
CO3. Demonstrate the physical problems using quantum computing.
CO4. Perform basic quantum computing operations.
CO5. Implement quantum computing to study CPS.

Syllabus

Quantum physics revisited, Introduction to Quantum Computation: Quantum bits, Bloch sphere representation of a qubit, multiple qubits.

Background Mathematics and Physics: Hilber space, Probabilities and measurements, entanglement, density operators and correlation, basics of quantum mechanics, Measurements in bases other than computational basis.

Quantum Circuits: single qubit gates, multiple qubit gates, design of quantum circuits. Quantum Information and Cryptography: Comparison between classical and quantum information theory. Bell states. Quantum teleportation. Quantum Cryptography, no cloning theorem. Quantum Algorithms: Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shorfactorization, Grover search, Noise and error correction: Graph states and codes, Quantum error correction, fault-tolerant computation.

. Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	M	M	H	L
CO2	H	M	H	L	M	H
CO3	H	H	H	H	M	M
CO4	M	H	M	L	H	L
CO5	M	M	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Nielsen M. A., Quantum Computation and Quantum Information, Cambridge University Press. 2002
2. Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific. 2004.
3. Pittenger A. O., An Introduction to Quantum Computing Algorithms, 2000.

23/7/25 Ruggi 24/7/25

Course Name:	Fault Detection and Applications	Contact hours	3+1+0
Course Code:	EE23403	Credit of the Course	4
Pre-requisite knowledge	Linear algebra, Computer programming skills, Control System Design, Nonlinear Control.		

Course Outcomes (COs)

At the end the course, students will be able to:

CO1. Study of fault detection problem in linear and nonlinear systems.
CO2. Analyse reliability and failure rates of a system against sensors/actuators faults.
CO3. Understand theoretical development and implementation of sliding mode schemes for fault tolerant control.
CO4. Analyse the robustness aspect of fault diagnosis and fault estimation of complex systems.
CO5. Apply fault detection schemes for a network of unmanned vehicles.

Syllabus

Module 1 Introduction

Fundamental concepts of fault, failure, detection, diagnosis, reliability, Redundancy, Coprime factorization techniques, Residual generator stability, Overview of fault detection (FD) and fault-tolerant control (FTC) technology.

Module 2 Fault Diagnosis and Fault Estimation

Conventional CCA, CCA based fault diagnosis method, DCCA-based fault detection for dynamic processes, The GRU-aided CCA fault detection method, Robust H_∞ fault detection for LDTV systems with multiplicative noise, Robust H_∞ fault detection for LDTV systems withmeasurement packet loss, Fixed-lag H_∞ fault estimator design for LDTV systems under an unreliable communication link.

Module 3 Robust Observer for Fault Detection and FTC

The Utkin observer, The Edwards–Spurgeon observer for fault reconstruction, Robust fault reconstruction using sliding mode observers, VTOL aircraft model, Observer variation, Comparisons with unknown input observers, Sliding Mode fault tolerant control (Actuator FTC, Sensor FTC).

Module 4 Fault diagnosis case studies: simulation and industrial applications

Geometric approach to FDI of linear and nonlinear systems, FDI problem formulation in a network of unmanned vehicles (centralized architecture, decentralized architecture, semi-decentralized architecture), Structured fault detection and isolation scheme, Actuator fault detection and isolation in a network of unmanned vehicles.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	L	L	L	L
CO2	H	L	L	H	M	L
CO3	H	H	M	M	M	L
CO4	M	M	H	M	L	L
CO5	M	M	H	M	M	H

[High: H; Moderately: M; Low: L]

References

1. Alwi, Halim, Christopher Edwards, and Chee Pin Tan. Fault detection and fault-tolerant control using sliding modes. London: Springer, 2011.
2. Martinez-Guerra, Rafael, and Juan Luis Mata-Machuca. Fault detection and diagnosis in nonlinear systems. Springer International Pu, 2016.
3. Karimi, Hamid Reza, ed. Fault Diagnosis and Prognosis Techniques for Complex Engineering Systems. Academic Press, 2021.
4. Meskin, Nader, and Khashayar Khorasani. Fault detection and isolation: Multi-vehicle unmanned systems. Springer Science & Business Media, 2011.

Course Name:	Network Control System	Contact hours	3+1+0
Course Code:	EE23304	Credit of the Course	4
Pre-requisite knowledge	Linear Algebra, Control System Design, Intelligent Control, Nonlinear Control.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Perform network analysis using graph theory.
CO2. Construct and analyze a discrete-time model for a dynamical system.
CO3. Analyze a multivariable dynamic system and design an appropriate controller for the system.
CO4. Evaluate the stability, performance and robustness of a closed-loop system.

Syllabus

Review of linear systems and Lyapunov stability, Features of control networks, Theory of nonnegative matrices (Perron–Frobenius theory).
Consensus Problem: cooperative control, leader-less and leader-follower architectures, formation control.
Linear matrix inequalities and switched systems, Stability analysis of networked control systems in presence of communication delays and packet loss/communication failure.
Multi-Agent System (MAS): Introduction, Different types of agents, Features and structure of MAS, Agent development software, Distributed optimization using multi-agent systems and Applications in real-time problems.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	M	H	H	L
CO2	M	L	L	M	H	M
CO3	H	M	M	H	M	L
CO4	H	M	H	H	M	H

[High: H; Moderately: M; Low: L]

References

1. M. Mesbahi and M. Egerstedt, Graph Theoretic Methods in Multiagent Networks, Princeton University Press, 2010.
2. L. Farina and S. Rinaldi, Positive Linear Systems, Wiley, 2000, ISBN: 978-0-471-38456-4.
3. M. Mesbahi and M. Egerstedt, "Graph Theoretic Methods in Multiagent Networks", Princeton Univercity Press, NJ, 2010.
4. W. Ren and R.W. Beard, "Distributed Consensus in Multi-vehicle Cooperative Control: Theory and Application", Springer-Verlag, London, 2008.
5. F. Lewis, H. Zhang, K. Hengster-Movric and A. Das, "Cooperative Control of Multi-Agent Systems: Optimal and Adaptive Design Approaches", Springer-verlag, London, 2014.
6. R. B. Bapat, "Graphs and Matrices", Hindustan Book Agency, Springer-Verlag, London, 2011.
7. L. Krick, M. E. Broucke and B. A. Francis "Stabilisation of infinitesimally rigid formations of multi-robot networks", International Journal of Control, Vol. 82, No. 3, 2009.
8. W. Ren, R. W. Beard, and E. M. Atkins, "Information Consensus in Multivehicle Cooperative Control", IEEE Control Systems Magazine, 2007.

Course Name:	Introduction to Digital Twin	Contact hours	3+1+0
Course Code:	EE23404	Credit of the Course	4
Pre-requisite knowledge	Internet-of-Things, Data Communication and Networking, Sensors and Interfacing.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Understand the industrial revolution and importance of digital twin.
CO2. Understand the architecture of digital twin.
CO3. Formulate the model and analyse the solutions of digital twin.
CO4. Formulate the technologies to create, manage and optimize digital twins.
CO5. Demonstrate the applications of digital-twin in engineering areas.

Syllabus

Introduction to digital twin (DT), Evolution of digital twin from computational techniques, Growth drivers for digital twin, Product & Process digital twins, Digital Model, Digital Shadow, Digital twin Prototype (DTP), Digital Twin Instance (DTI), Digital Twin Aggregate (DTA), Partial digital twin, Clone digital twin, augmented digital twin, Different tools in digital twin.

DT in dynamical systems: solutions of ODE & PDE, Time-domain and frequency domain simulations of dynamical systems with DT, Extensive analysis of Time-domain and frequency domain simulations, Data-driven DT.

Bayesian methods in digital twin technology – Kalman filter based approaches.

Digital twin for multi-timescale dynamical systems - Mixture of Gaussian Process and Expectation maximization.

Digital twin from multi-fidelity data, Digital twin for systems with partially known physics, Enabling technologies of digital twin: AI, ML/DL, etc.

Applications of digital twin: product development, smart cities, smart grid, Health care, etc.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	M	H	L
CO2	H	M	M	L	M	L
CO3	M	H	M	L	M	L
CO4	M	H	L	M	H	L
CO5	H	H	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Anand Iyer, Digital Twin: Possibilities of the new Digital twin technology, 2017.
2. A. Iserles, A first course in the numerical analysis of differential equations (No. 44). Cambridge university press, 2009.
3. C.M. Bishop, Pattern recognition and Machine learning, Springer, 2007.
4. K.P. Murphy, Machine learning: A Probabilistic Perspective, MIT press, 2012.
5. C.E. Rasmussen, C.K.I. Williams, Gaussian Processes for Machine Learning, MIT Press, 2006.

Course Name:	Edge and Cloud Computing	Contact hours	3+1+0
Course Code:	EE23405	Credit of the Course	4
Pre-requisite knowledge	Internet-of-Things, Data Communication and Networking.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Understand the concept of cloud and edge computing.
CO2. Demonstrate the architecture of edge computing.
CO3. Perform data management applying edge computing.
CO4. Apply the concept of edge computing to address the industrial problems.
CO5. Apply advanced computation algorithms to support latency-based applications.

Syllabus

Introduction to cloud computing, Architecture, Service Management in Cloud Computing: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service(SaaS), Data Management in Cloud Computing, Resource Management in Cloud Computing, Cloud Implementation, Fog Computing, VM Resource Allocation, Management and Monitoring.

Limitations of cloud computing to support low latency and RTT (round-trip time), Edge computing: Introduction and Architecture, Concepts of distributed systems in edge computing, time ordering and clock synchronization, and distributed snapshot, Serverless Computing and FaaS Model, Cloud-Fog-Edge enabled Analytics, Cloud Security.

Introduction to MQTT and Kafka for end-to-end IoT pipeline, Edge computing applications: image classifier, self-driving cars, etc.

Deep Learning On-Device inference at the edge to support latency-based application.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	H	M	L	H	L
CO2	H	H	H	L	H	L
CO3	L	L	M	L	H	M
CO4	H	M	H	M	H	H
CO5	M	M	H	L	H	M

[High: H; Moderately: M; Low: L]

References

1. Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Cloud Computing: Principles and Paradigms, Wiley, 2011.
2. Gautam Shroff, Enterprise Cloud Computing - Technology, Architecture, Applications, Cambridge University Press, 2010.
3. Barrie Sosinsky, Cloud Computing Bible, Wiley-India, 2010.
4. Nikos Antonopoulos, Lee Gillam, Cloud Computing: Principles, Systems and Applications, Springer, 2012.
5. Rajkumar Buyya, Satish Narayana Srirama, Fog and Edge Computing: Principles and Paradigms, Wiley, 2019.
6. Rajiv Misra, Yashwant Patel, Cloud and Distributed Computing: Algorithms and Systems, Wiley 2020.

Course Name:	State-of-Art Seminar	Contact hours	0+0+6
Course Code:	EE23654	Credit of the Course	4
Pre-requisite knowledge	Course Works.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Explore and select relevant topic(s) of their research work.
CO2. Assimilate literature on reported works of chosen topic(s)/area.
CO3. Find the research gaps in the defined/selected area.
CO4. Formulate the methodologies to address the identified research problem(s) through state-of-art review.
CO5. Write and present the work in the department.

Syllabus

Every student admitted to the candidacy for the M.Tech. degree specializing in "Cyber Physical Systems" is required to give a State-of-Art Seminar covering the area of research in line with the proposed programme. ***There is no specific syllabus for this course. Student can identify and select any topic, of his / her interest, pertaining to Cyber Physical Systems.***

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	M	L	H
CO2	H	M	H	H	H	H
CO3	H	M	M	H	M	M
CO4	H	M	H	H	M	H
CO5	M	H	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Books/Journals/Conference proceedings.

Course Name:	Thesis-I	Contact hours	0+0+18
Course Code:	EE23604	Credit of the Course	12
Pre-requisite knowledge	Course Works.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Demonstrate the research problem(s) through rigorous review of State-of-Art works.
CO2. Find the research gaps and establish scope of works.
CO3. Develop methodology for the study.
CO4. Perform theoretical/real-time assessment.
CO5. Write and present the work in the department.

Syllabus

There is no specific syllabus for this course. Student can identify and select any topic, of his / her interest, pertaining to Cyber Physical Systems.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	H	M	M	M	L	M
CO2	M	M	H	M	L	L
CO3	H	H	H	H	H	M
CO4	H	H	H	H	M	H
CO5	M	H	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Books/Journals/Conference proceedings.

Semester-IV

Course Name:	Thesis-II	Contact hours	0+0+30
Course Code:	EE24604	Credit of the Course	20
Pre-requisite knowledge	Course Works.		

Course Outcomes:

At the end the course, students will be able to:

CO1. Expand the defined research problem(s) statement.
CO2. Perform in-depth analysis of the developed methodology through analytically/experimentally.
CO3. Analyze the measurements/data and offer solutions.
CO4. Frame concluding remarks of their work and develop products.
CO5. Write and present the work in the department.

Syllabus

There is no specific syllabus for this course. Student can identify and select any topic, of his / her interest, pertaining to Cyber Physical Systems.

Mapping between COs and POs

Course Outcomes (COs)	Programme Outcomes (POs)					
	PO-1	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**
CO1	M	M	M	L	L	M
CO2	M	H	H	H	M	H
CO3	H	H	H	M	M	H
CO4	M	H	M	M	H	H
CO5	M	H	H	M	H	H

[High: H; Moderately: M; Low: L]

References

1. Books/Journals/Conference proceedings.